Kuan-Chen Chen

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EDUCATION

University of Florida **Gainesville, FL**
Master of Science in Applied Data Science *8/2024 - 5/2026*
National Sun Yat-Sen University **Kaohsiung, Taiwan**
Bachelor of Science in Applied Mathematics *9/2019 - 6/2023*
***Honors & Awards*:** Valedictorian, Department of Applied Mathematics; Distinguished Service Award

EXPERIENCE

NVIDIA NeMo Automodel (MiniCPM5-1B, Falcon H1)
Open-Source Contributor *5/2026-Present*
- Contributed 2 merged PRs to NVIDIA's distributed LLM training library, adding 7 fine-tuning recipes (SFT, LoRA) for models from 0.5B to 34B
- Configured LoRA hyperparameters (rank, alpha, target modules) and tensor parallelism for 34B-scale training; validated convergence on T4 GPU runs

Florida Gators Women's Basketball, University of Florida **Gainesville, FL**
Data Scientist Intern *8/2025 – 5/2026*
- Replaced a manually-weighted player evaluation system with OA-Rating, an XGBoost + SHAP model validated against the legacy system (r = 0.78) across 120K+ observations, 5,500+ players, 350+ teams
- Built Elo-based team rating system with 83% agreement with NCAA's official national rankings, and clustered teams into 4 tactical archetypes to support real-time game preparation and opponent scouting
- Shipped production R Shiny dashboard that contributed to team improvement from 60th nationally (16–17) to 48th nationally (18–15)

Bioinformatics Lab, University of Florida **Gainesville, FL**
Student Research Assistant *12/2024 – 5/2026*
- Designed stacking ensemble reaching 87.6% prediction accuracy (ROC-AUC 0.876) in distinguishing brain- vs. blood-derived sequences, outperforming the best individual model (85%) on 10,486 CDR3 sequences
- Developed biologically-informed data augmentation algorithm using BLOSUM62 substitution matrix, generating conservative sequence variants to expand minority class 10× (432 → 4,622) and mitigate severe class imbalance (1:23 → 1:1.5)
- Engineered 42 physicochemical features from protein sequences and optimized decision thresholds under class imbalance, improving precision to 0.716 for high-confidence, interpretable predictions

PROJECTS

CV Generator (JD-Tailored Résumé Agent) Ckcinnabar/CV Generator **Gainesville, FL**
- Built a hybrid retrieval pipeline — dense embeddings (Chroma) + BM25, fused via Reciprocal Rank Fusion — with a Claude Haiku selector ranking the top-5 most relevant items per JD
- Orchestrated via a LangGraph stateful graph with a human-in-the-loop interrupt for approval, and constrained Claude generation with hard-coded titles/dates to prevent hallucination
- Built an LLM supervisor that fact-checks every claim against source documents, plus a render-retry loop (docx2pdf → pypdf) that shortens content until it fits one page

cAIuldron (Recipe Generation LLM) Ckcinnabar/cAIuldron **Gainesville, FL**
- Built an AI recipe app that turns ingredient photos into complete recipes with AI-generated images; shipped two production versions, cutting pipeline latency from 40–80s to 10–20s
- Developed v1 pipeline using CLIP and DETR computer vision models for multi-ingredient detection across 525+ food classes (88% F1); fine-tuned Llama 3.2 1B via QLoRA on 7,913 curated RecipeNLG recipes, updating only 0.23% of parameters (2.8M) while achieving 92/100 recipe quality score
- Rebuilt v2 on LangGraph with Groq-hosted Llama 4 Scout (vision) and Llama 3.3 70B (generation); parallel fork/join ChromaDB RAG + Tavily search, removing the 4–6 GB VRAM dependency

SKILLS

Languages: Python, SQL, R, JavaScript, TypeScript
Machine Learning: Deep Learning, NLP, Feature Engineering, PyTorch, scikit-learn, XGBoost, LightGBM, SHAP
LLM & GenAI: Fine-Tuning (LoRA/QLoRA), RAG, HuggingFace Transformers, LangChain, LangGraph, LangSmith
Cloud & DevOps: AWS, GCP, Docker, CI/CD (GitHub Actions), Git
Web & Apps: FastAPI, Flask, React.js, Gradio, Dash, R Shiny
Data & Viz: pandas, NumPy, MySQL, PostgreSQL, MongoDB, ChromaDB, Power BI, Tableau
Certifications: NVIDIA Fundamentals of Deep Learning; IBM Data Science Professional Certificate